

Homework 5

Problem 1. Let $M = B(0, 1)$ in $\mathbb{R}^3$ and $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a smooth (infinitely differentiable) vector field. Let $\vec{n}$ be the outward unit vector on $\partial M = S^2$ (the 2-dimensional sphere).

1. Show that for any open cover $M \subset \cup_i B_i$ we can split $F = \sum_{j \in J} F_j$ on M where for each $j \in J$, there is $i \in I$ such that $\text{supp} F_j \subseteq B_i$.
2. Show that for any smooth vector field $G : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with compact support we have

$$\iiint_{\mathbb{R}^3_+} \text{div} G = \iint_{\partial \mathbb{R}^3_+} G \cdot (-\vec{e}_3) dS$$

3. Show the divergence theorem with spherical parametrization

$$\iiint_M \text{div} F = \iint_{\partial M} (F \cdot \vec{n}) dS$$

where $\vec{n}$ is the outwards unit normal vector.

Hint: use spherical coordinates $W(r, \theta, \phi) = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta)$ to parametrize around $W(1, \frac{\pi}{2}, 0) = (1, 0, 0)$. Here $\{r = 1\}$ corresponds to the boundary and $\frac{\partial_r W}{|\partial_r W|}$ is the outwards unit normal vector. Observe that $|\partial_\theta W \times \partial_\phi W| = |\det DW| = |r^2 \sin \theta|$. $\text{div} F$ when translated to spherical coordinates becomes $\frac{1}{r^2} \partial_r (r^2 \tilde{F}_r) + \frac{1}{r \sin \theta} \partial_\theta (\sin \theta \tilde{F}_\theta) + \frac{1}{r \sin \theta} \partial_\phi (\tilde{F}_\phi)$ where $\tilde{F}_r = (F \circ W) \cdot \partial_r W$, and similarly for $\tilde{F}_\theta$ and $\tilde{F}_\phi$. So

$$(\text{div} F)(\text{Jacobian}) = \partial_r (r^2 \tilde{F}_r \sin \theta) + \partial_\theta (r \sin \theta \tilde{F}_\theta) + \partial_\phi (r \tilde{F}_\phi).$$

Problem 2. Write the full details of the inverse function theorem from lecture (use problem 3 of midterm).

Let $f : U \rightarrow V$ be C^k where U, V are open in $\mathbb{R}^n$, $k \geq 1$ and $p \in U$.

1. Show that there are open sets U_0 containing p and V_0 containing $f(p)$ such that $f|_{U_0}^{V_0} : U_0 \rightarrow V_0$ is invertible.
2. Also show that $g := (f|_{U_0}^{V_0})^{-1}$ is C^k (not proved in lecture).

Hint: to keep notation readable, try to argue that WLOG $p = 0, f(0) = 0, f'(0) = \text{Id}$. You can not directly use the chain rule (g not differentiable yet), but the chain rule on differentiating $g(f(x)) = x$ should indicate the candidate for g' at $f(x)$. Then it is a matter of proving it is a genuine Frechet derivative. To keep notation readable, let $\tilde{x} = f(x)$ and introduce h and $\tilde{h}$ such that $f(x + h) = \tilde{x} + \tilde{h}$. After showing g' exists, show that the formula for g' is C^{k-1} .

Problem 3. Connect divergence theorem to generalized Stokes' theorem.

Let $M = B(0, 1)$ in $\mathbb{R}^3$ for simplicity. Define the differential form

$$\omega = f_1(x) dx_2 \wedge dx_3 - f_2(x) dx_1 \wedge dx_3 + f_3(x) dx_1 \wedge dx_2$$

where $F := (f_1, f_2, f_3)$ is a smooth vector field. Show that

$$\begin{aligned} \int_{\partial M} (F \cdot \vec{n}) dS &= \int_{\partial M} \omega \\ \int_M \operatorname{div} F &= \int_M d\omega \end{aligned}$$

where d is the exterior derivative, $\vec{n}$ is the outwards unit normal vector, ∂M is locally parametrized (and oriented) by $T(y_1, y_2) = (x_1, x_2, x_3)$ such that $\vec{n} = \frac{\partial_1 T \times \partial_2 T}{|\partial_1 T \times \partial_2 T|}$.

Hint: Partition of unity first to assume F is only supported near a point $p \in \partial M$ (becoming 0 outside). Also recall that for any vectors u, v, w in $\mathbb{R}^3$:

$$u \cdot (v \times w) = \det \begin{pmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{pmatrix}$$